Individual Report

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Group Number: 27
Project Title: MarketSense AI: A Neuro-Symbolic Intelligent Reasoning Simulator for Macroeconomic Shocks & Portfolio Adaptability

My contributions to the project

As the sole architect and developer of **MarketSense AI**, I conceptualized, designed, and implemented the full-stack platform end-to-end:

- **Deterministic Financial Core:** Engineered an execution engine covering closed-form pricing models, STCG/LTCG tax drag calculations, 0.15% brokerage friction, dynamic circuit breakers, and institutional benchmarking—ensuring 100% mathematical reproducibility across simulations.
- **Knowledge Graph & GraphRAG Pipeline:** Constructed a 36-node, 48-edge macroeconomic causal graph using NetworkX. Integrated this with a hybrid RAG architecture backed by ChromaDB (indexing 52 curated domain chunks across 6 legendary investors and 22 crisis scenarios) to ground model reasoning in deterministic causal links.
- **Dual-Runtime Socratic AI Mentor:** Orchestrated an adaptive reasoning agent operable in dual modes: a cloud-hosted Gemini 2.5 Flash runtime and an air-gapped, offline Ollama environment for private inference.
- **Interactive Full-Stack Interface:** Built a reactive Streamlit frontend featuring Pyvis physics-driven causal graph visualizers, a counterfactual market sandbox, and end-to-end deployment on Streamlit Community Cloud.

Self-reflections

This project deepened my conviction that AI must serve society transparently. To combat wealth destruction among retail investors during financial crises, I channeled intelligent reasoning into an educational flight simulator MVP. Transforming theoretical coursework into a commercial 3-tier SaaS product demonstrated how ethical, neuro-symbolic AI can empower society, fostering disciplined financial literacy and emotional resilience safely before individuals risk life savings.

My most impressive achievement was thinking like a product designer, engineer, and customer all at once. Applying concepts taken away from classes helped me realize it without question. It has opened my thinking to manage AI-driven features in an app rather than offloading all features only to AI. I managed UI experience with Fast and slow cognitive architecture, used KG and GraphRAG for explanation, and integrated NLP to assign personality to financial mentoring. I feel confident with applying my learnings into this practice module project and from GC1, has made move closer to my goal of becoming AI Engineer/Architect building responsibly and explainable AI Apps.